MIL-T-8025A(USAF)
7 July 1969
SUPERSEDING
MIL-T-8025(USAF)
3 December 1952

MILITARY SPECIFICATION

TRANSMITTER, PRESSURE, LOG ABSOLUTE, TYPE MC-1

1. SCOPE

1.1 This specification covers one type of log absolute, pressure transmitter, designated type MC-1.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods Of
MIL-D-1000	Drawings, Engineering And Associated List
MIL-C-5015	Connectors, Electric, "AN" Type
MIL-E-5272	Environmental Testing, Aeronautical And Associated Equipment, General Specification For
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements For
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification For
MIL-A-8625	Anodic-Film; Corrosion-Protective (For) Aluminum Alloys

FSC 6685

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STANDARDS

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-143	Specification And Standards, Order Of Precedence For The Selection Of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, And Waterproofing; With Appropriate Test Methods
MS27572	Case, Instrument, Flange-Mounted, Aircraft
MS33678	Connector, Receptacle, Electrical, Integral Mounting

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules And Regulations

(Applications for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago IL 60606.)

3. REQUIREMENTS

* 3.1 Qualification testing. This specification makes provisions for qualification testing.

* 3.2 Qualification. Transmitter furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.4).

* 3.3 Materials.

* 3.3.1 Materials shall conform to applicable specifications as specified herein. Materials that are not specifically designated shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

* 3.3.2 Metals. Metals shall be of the corrosion-resistant type or shall be suitably protected to resist corrosion during normal service life.

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* 3.3.3 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the transmitter, except where magnetic materials are essential.

* 3.3.4 Protective treatment. When materials are used in the construction of the transmitter that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

* 3.3.4.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except that the dial, small holes, and case inserts need not be anodized.

* 3.3.4.2 Steel parts. Steel parts, where practicable, shall be cadmium plated in accordance with QQ-P-416, Type I or II, as applicable, and of a class that is adequate to achieve the degree of protection required.

* 3.3.5 Selection of specifications and standards. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in the following paragraph.

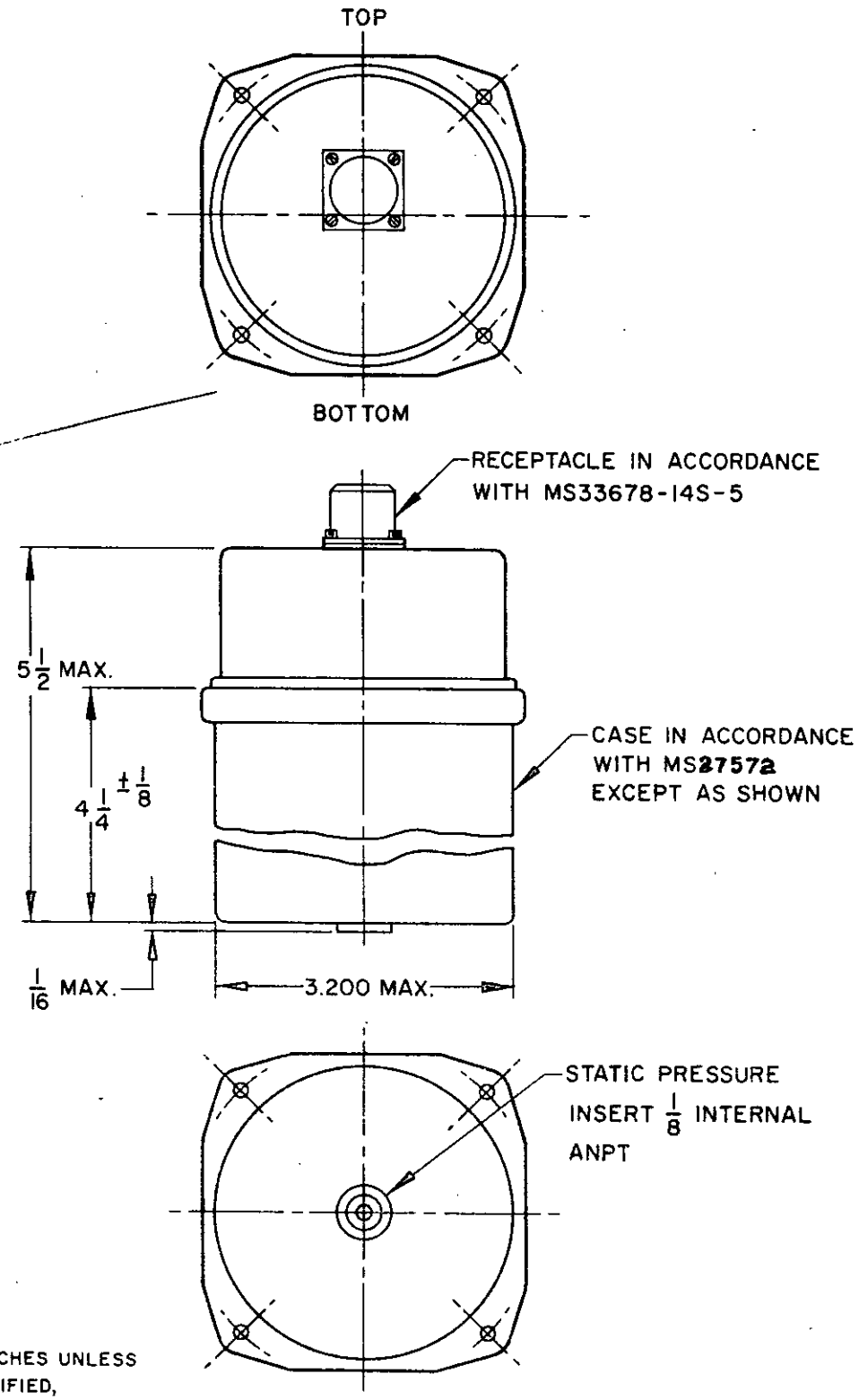
* 3.3.5.1 Standard parts. AN or MIL Standard parts shall be used wherever they are suitable for the purpose, and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, et cetera, may be used, provided they have suitable properties and are replaceable by the AN or MIL Standard parts without alteration, and provided the corresponding AN or MIL part numbers are referenced on the drawings and in the parts lists. In applications for which no suitable corresponding AN or MIL part is in effect on date of invitation for bids, commercial parts may be used provided they conform to the requirements of this specification.

* 3.4 Design and construction. The transmitter shall be substantially as shown on Figure 1. The transmitter shall be constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service without failure.

* 3.4.1 The transmitter shall consist of a static pressure operated altimeter mechanism operating a synchro control transformer whose electrical characteristics conform to 3.4.2.

* 3.4.1.1 The transmitter shall be designed so that proper compensation is made for variation in ambient temperature.

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DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED,
TOLERANCES:
FRACTIONS ± 1/6

FIGURE 1 HOUSING

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* 3.4.2 Synchro-control transformer. The synchro-control transformer shall be a 1-phase, 3-phase control transformer. The input impedance of the 3-phase winding (line-to-line) shall be 25 ohms or greater at 400 cycles. When the 3-phase winding is excited by 5.9 volts (line-to-line maximum) the output voltage per degree of error signal shall be at least 0.10 volt on open circuit. The resistance load across the output required to reduce open circuit output voltage to 1/2 its value shall be 3800 ohms or less at 25° Centigrade (C).

* 3.4.2.1 The synchro shall be connected to the altimeter mechanism so that the angular rotation of the synchro rotor shall turn as a function of the following equation:

$$\phi - 173.56 \log_{10} \frac{P}{29.92}$$

(Symbol ϕ signifies angle of rotation of the synchro transmitter in degrees; P indicates static pressure in inches mercury.) Suitable stops shall be provided to prevent movement of the synchro rotor at altitudes above 80,000 feet.

* 3.4.3 Housing. The transmitter housing shall consist of a case and a cap, and shall be in accordance with Figure 1.

* 3.4.3.1 Case. The case shall be made either of nonferrous low density metal or of synthetic material, uniform in texture with a smooth surface, and entirely covered with a durable black finish. Synthetic material shall be of the thermal setting type composed of a suitable filler and phenol condensation product binder.

* 3.4.3.2 Reinforcement. The inside of the housing around each connection shall be of sufficient strength to prevent damage to the housing when the connections are tightened during installation of the transmitter.

* 3.5 Performance. The transmitter shall be capable of meeting the requirements specified in Section 4 when subjected to the applicable tests.

* 3.6 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-1000.

* 3.7 Electrical connections. The electrical connector used in the construction of the transmitter shall be in accordance with MIL-C-5015.

* 3.7.1 The pins of the electrical connector shall be connected to the synchro as follows:

Pin A - Single Phase Winding No. 1
Pin B - Single Phase Winding No. 2
Pin C - Three Phase Winding No. 1
Pin D - Three Phase Winding No. 2
Pin E - Three Phase Winding No. 3

The phase rotation shall be C-D-E for increasing pressure.

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* 3.7.2 Electrical zero. The electrical zero of the transmitter shall be set at zero altitude (29.92 inches mercury).

* 3.8 Threads.

* 3.8.1 Screw. Unless otherwise specified, the threads of all machine screws, 0.060 inch or larger in diameter, shall conform to MIL-S-7742.

* 3.8.2 Pipe. Pipe threads shall conform to MIL-P-7105.

* 3.9 Weight. The weight of the transmitter completely assembled, less mounting screws, shall not exceed 1.5 pounds.

* 3.10 Marking.

* 3.10.1 The exterior of the housing shall be finished in durable dull black, except as specified herein.

* 3.10.2 The back of the case, adjacent to the pressure connections, shall be marked as follows:

"S" Static Connection

* 3.10.2.1 The following shall also be marked on the back of the case:

CAUTION
DO NOT BLOW IN TUBE

* 3.11 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

* 3.12 Installation.

* 3.12.1 Mounting screws. The contractor shall furnish sufficient mounting screws for installing the transmitter. They shall be No. 6-32 National Coarse (Thread) NC-2 round head brass machine screws having a durable dull black finish. The length shall be sufficient to permit mounting on a panel 3/16 inch thick.

* 3.12.2 Envelope. An envelope containing the mounting screws shall be furnished with each transmitter. The following information shall be printed on the face of the envelope:

IMPORTANT
THIS ENVELOPE CONTAINS
MOUNTING SCREWS

* 3.13 Workmanship.

* 3.13.1 General. The transmitter, including all parts and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall

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be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, painting, wiring, and riveting; alignment of parts and tightness of assembly screws and bolts, et cetera.

* 3.13.2 Cleaning. The transmitter shall be thoroughly cleaned, and loose, spattered, or excess solder, metal chips and other foreign material removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of tests. The inspection and testing of the transmitter shall be classified as follows:

a. Qualification tests (see 4.4).

b. Quality conformance inspection (see 4.5).

* 4.3 Test conditions.

* 4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of test are not specified definitely, it is understood that the test is to be made at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

* 4.3.2 Tapping. Unless otherwise specified, the transmitter shall be gently tapped or vibrated before a test reading is taken.

* 4.3.3 Attitude. Unless otherwise specified, the transmitter shall be tested in normal operation position. Normal operation position shall be with mounting flange in a vertical plane and the top side up.

* 4.3.4 Vibration. A vibration stand shall be used which will vibrate at any desired frequency between 500 and 3000 cycles per minute and shall subject the transmitter to vibration such that a point of the case will describe in a plane inclined 45 degrees to the horizontal plane, a circle, the diameter of which is equal to the amplitude specified herein.

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* 4.3.5 Unless otherwise specified, whenever indications from the transmitter are required, the synchro control transformer shall be connected to a follow-up synchro through an amplifier and servo system as shown on Figure 2. Facilities for determining the rotor position of the follow-up synchro to within ±0.1 degree shall be provided and the servo system shall not introduce more than 0.1 degree error into the indication. The single phase winding of the follow-up synchro shall be excited with 26 volts, 400 cycle current and the signal for the amplifier shall be obtained from the single phase winding of the synchro on the transmitter.

* 4.3.6 Unless otherwise specified, the test tolerances shall be as specified in Tables I and II.

* 4.4 Qualification test.

* 4.4.1 Prior qualification. Unless otherwise specified by the procuring agency, transmitters which have not previously passed a qualification test, or which have passed the qualification test and have been modified in any manner, shall satisfactorily pass a qualification test prior to the acceptance of any transmitter.

* TABLE I

Altitude Feet	Pressure Millimeter of Mercury	Indication Degrees	Tolerance Degrees
-1,000	787.9	2.7	2.0
* 0	760.0	0.0	2.0
5,000	662.3	346.1	2.0
*10,000	522.6	331.8	3.0
15,000	428.8	316.9	3.0
*20,000	349.1	301.4	3.0
25,000	281.9	285.3	3.0
*30,000	225.6	268.4	3.0
35,000	178.7	250.9	3.0
*40,000	140.7	232.9	3.0
45,000	110.8	214.8	3.0
50,000	87.30	196.9	3.0
55,000	68.76	178.9	4.0
*60,000	54.15	160.9	4.0
65,000	42.65	142.9	4.0
70,000	33.59	124.9	4.0
75,000	26.45	106.9	4.0
80,000	20.83	88.9	4.0

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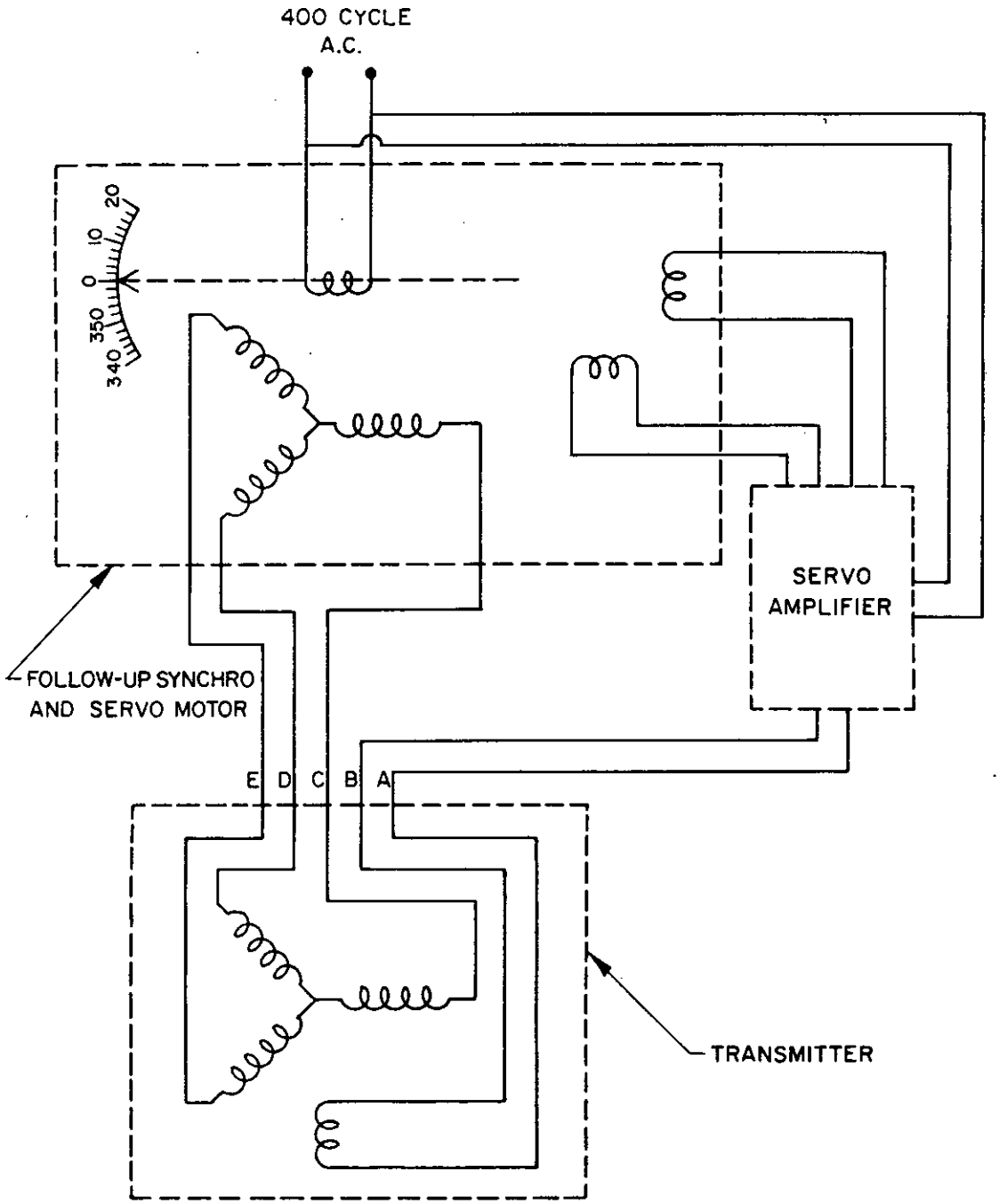


FIGURE 2. TRANSFORMER CONNECTION

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TABLE II

TESTS	TOLERANCES (±) AND TEST POINTS
Hysteresis	
First test point	30,000 Feet
Tolerance	2.0 Degrees
Second test point	15,000 Feet
Tolerance	2.0 Degrees
Third test point	Atmospheric
Tolerance	2.0 Degrees
Friction	
Tolerance	2.0 Degrees
Position error	
Tolerance	2.0 Degrees
Vibration error	10,000 Feet
Test pressure, altitude	30,000 Feet
Null Signal amplitude increase at 10,000 and 30,000 feet	0.2 Volt
Null shift tolerances at 10,000 and 30,000 feet	2.0 Degrees
Vibration failure	
Test pressure altitude	30,000 Feet
Tolerance	2.0 Degrees
Seasoning	
Tolerance, Scale error	2.0 Degrees
Low temperature	
Tolerance, Scale error	2.0 Degrees
High temperature	
Tolerance, Scale error	2.0 Degrees

* 4.4.2 Qualification test samples. Qualification test samples shall consist of 3 transmitters. Samples shall be appropriately identified with the manufacturer's own part number and any additional identification required by the authorizing letter.

* 4.4.3 The qualification test shall consist of all the tests specified herein.

* 4.5 Quality conformance inspection. Quality conformance inspection shall consist of the following:

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a. Individual tests (see 4.6).

b. Sampling tests (see 4.7).

* 4.6 Individual tests. Each transmitter shall be subjected to the following tests.

* 4.6.1 Examination of product. Each transmitter shall be inspected to determine compliance with this specification with respect to material, workmanship, and envelope.

* 4.6.1.1 Dielectric strength. A potential of 500 volts alternating current (AC) at commercial frequency shall be applied between each pin and the case of the transmitter for a minimum period of 5 seconds. There shall be no breakdown of the insulation.

* 4.6.1.2 Electrical zero. The transmitter shall be tested for electrical zero as follows:

* 4.6.1.3 Leads from pins C, E, and A shall be tied together. A 0.1 microfarad condenser shall be connected across leads from pins A and B. A vacuum tube voltmeter shall be connected between leads from pins B and D. A voltage of 9 volts, 400 cycles shall be applied between pins C and D. With the connections specified in 4.6.1.4.1, pressure altitude shall be adjusted to 5,000 feet. Pressure shall be slowly increased (altitude decreased). The voltage should increase as the pressure is increased (altitude decreased) from 5,000 to 1,000 feet. If the voltage increase, the voltmeter lead shall be disconnected from pin D and connected to pin A. The pressure shall be increased (altitude decreased) until the voltmeter indicates the minimum voltage. This is electrical zero. The tolerances specified in Table I shall apply.

CAUTION: Do not exceed 31.02 inches mercury in this test.

* 4.6.1.4 Scale error. The transmitter shall be mounted on a panel that has facilities for tapping to remove the friction effect.

* 4.6.1.4.1 The static pressure connection of the transmitter shall be connected by means of a tee to an accurate barometer and to a source of air pressure and suction.

* 4.6.1.4.2 The electrical connection shall be as specified in 4.3.5.

* 4.6.1.4.3 The scale error test shall be made by successively subjecting the transmitter to the pressures marked by an asterisk in Table I. The reduction in pressure, as indicated by a standard mercurial barometer during the interval between test points, shall be made at a rate corresponding to an increase in altitude of approximately 6,000 feet per minute. The transmitter shall remain at the pressure corresponding to each test point for at least 30 seconds but not more than 5 minutes before a test reading is taken. The errors at the test points shall not exceed the tolerances listed in Table I. Movement of the indicator shall be free from irregular motion when the pressure is varied

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uniformly. When desired by the Inspector, the transmitter may be tested for scale error at any additional point or points on the scale as listed in Table I. The error shall not exceed the tolerances listed in the Table.

* 4.6.1.5 Hysteresis. Not more than 15 minutes after the transmitter has been first subjected to the pressure corresponding to the upper limit of the scale specified in 4.6.1.4, the pressure shall be increased at a rate corresponding to a decrease in altitude of approximately 6,000 feet per minute until the pressure corresponding to the first test point given in Table II is reached. The transmitter shall remain at this pressure for at least 5 minutes but not more than 15 minutes before the test reading is taken. After the reading has been taken, the pressure shall be further increased at the above rate until reached. The transmitter shall remain at this pressure for at least one minute but not more than 10 minutes before a test reading is taken. After the reading has been taken, the pressure shall be further increased at the above rate until atmospheric pressure is reached. The reading of the indicator at either of the 2 test points shall not differ from the reading of the indicator for the corresponding altitude in the scale error test by more than the tolerances given in Table II.

* 4.6.1.6 After effect. Not less than one minute and not more than 5 minutes after the completion of the test specified in 4.6.1.5, the indicator shall have returned to its original indication, corrected for any change in atmospheric pressure, within the tolerances given in Table II as the third test point.

* 4.6.1.7 Friction. The transmitter shall be tested for friction at every point indicated by an asterisk in Table I. The pressure shall be decreased to bring the pointer approximately to the desired reading and then held constant while 2 readings are taken; the first before the transmitter is tapped, and the second after the transmitter is tapped. The difference between any 2 such readings shall not exceed the tolerances specified in Table II. The pointer shall move smoothly while the pressure is varied uniformly without vibration of the transmitter. This test may be combined with the test specified in 4.6.1.4. During this test, the servo indicator may be tapped to remove any friction in the indicator and servo system.

* 4.6.1.8 Leakage.

* 4.6.1.8.1 Case leak. The static pressure connection shall be connected to the mercury manometer and a source of suction. A suction of 15 inches mercury shall be applied to the static pressure connection of the transmitter. With the source disconnected during a period of one minute, the difference between the mercury levels in the manometer shall not change by more than 0.2 inch.

* 4.7 Sampling tests. Five transmitters shall be selected at random from each lot and subjected to the following tests:

* 4.7.1 Lot. A lot for inspection purposes shall consist of transmitters manufactured under the same conditions, submitted for inspection at substantially the same time and shall not exceed 500 units.

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* 4.7.2 Individual units failing to meet their respective tests shall be rejected. When tests are specified on a quantity of units that are selected as representative of a certain lot, and one or more of this number fail to meet the specified tests, the lot shall be held in abeyance until the extent and cause of failure is determined. Individual performance tests need not be interrupted unless the defect is of such a nature that it will seriously affect the performance or safe use of the equipment, in which case the lot shall be rejected. Rejected units shall be replaced or shall be reworked to correct the defects, after which all necessary tests shall be repeated.

* 4.7.3 Weight. The transmitter shall be weighed completely assembled. The weight of the transmitter shall not exceed 1.5 pounds.

* 4.7.4 Magnetic property. The magnetic property of the transmitter shall be determined in terms of the deflection of a free magnet, approximately 1-1/2 inches long, in a magnetic field with a horizontal intensity of 0.18 ±0.001 oersted, with the transmitter held in various positions on an East-West line 5 inches from the center of the magnet. (An aircraft compass with compensating magnets removed therefrom, may be used as the free magnet for this test.) The maximum deflection of the magnet shall not exceed one degree. This test shall be conducted with electrical power connected to the synchro.

* 4.7.5 Low temperature operation. The transmitter shall be placed in a chamber and maintained at a temperature of -54 ±2°C for a period of 4 hours with no electrical power applied. While the transmitter is still at this temperature, power shall be applied and the transmitter shall meet the requirements specified for scale errors as specified in 4.6.1.4. The errors at the test points shall not exceed the tolerances specified in Table I by more than the tolerances specified in Table II.

* 4.7.6 High temperature operation. The transmitter shall be placed in a chamber and maintained at a temperature of 71 ±2°C for a period of 4 hours with no electrical power applied. While the transmitter is still at this temperature, power shall be applied and the transmitter shall meet the requirements specified for scale error as specified in 4.6.1.4. The errors at the test points shall not exceed the tolerances specified in Table I by more than the tolerances specified in Table II.

* 4.7.7 Seasoning. The transmitter shall be subjected to 1,000 applications of an absolute pressure of approximately 20 inches mercury. These applications shall be made at a rate not greater than 6 per minute. Not less than one hour following this test, the transmitter shall be tested for scale errors as specified in 4.6.1.4. The errors at the test points shall not exceed the tolerances specified in Table I by more than the tolerances specified in Table II.

* 4.7.8 Vibration error. This test shall be conducted in accordance with vibration tests, Procedure IV of MIL-E-5272. This test shall be conducted while the absolute pressures specified in Table II are applied to the indicator. Electrical connections shall be made to the transmitter as shown on Figure 3 and the standard synchro rotated to the null voltage as shown by the lowest amplitude signal on the oscilloscope. The transmitter shall be subjected to

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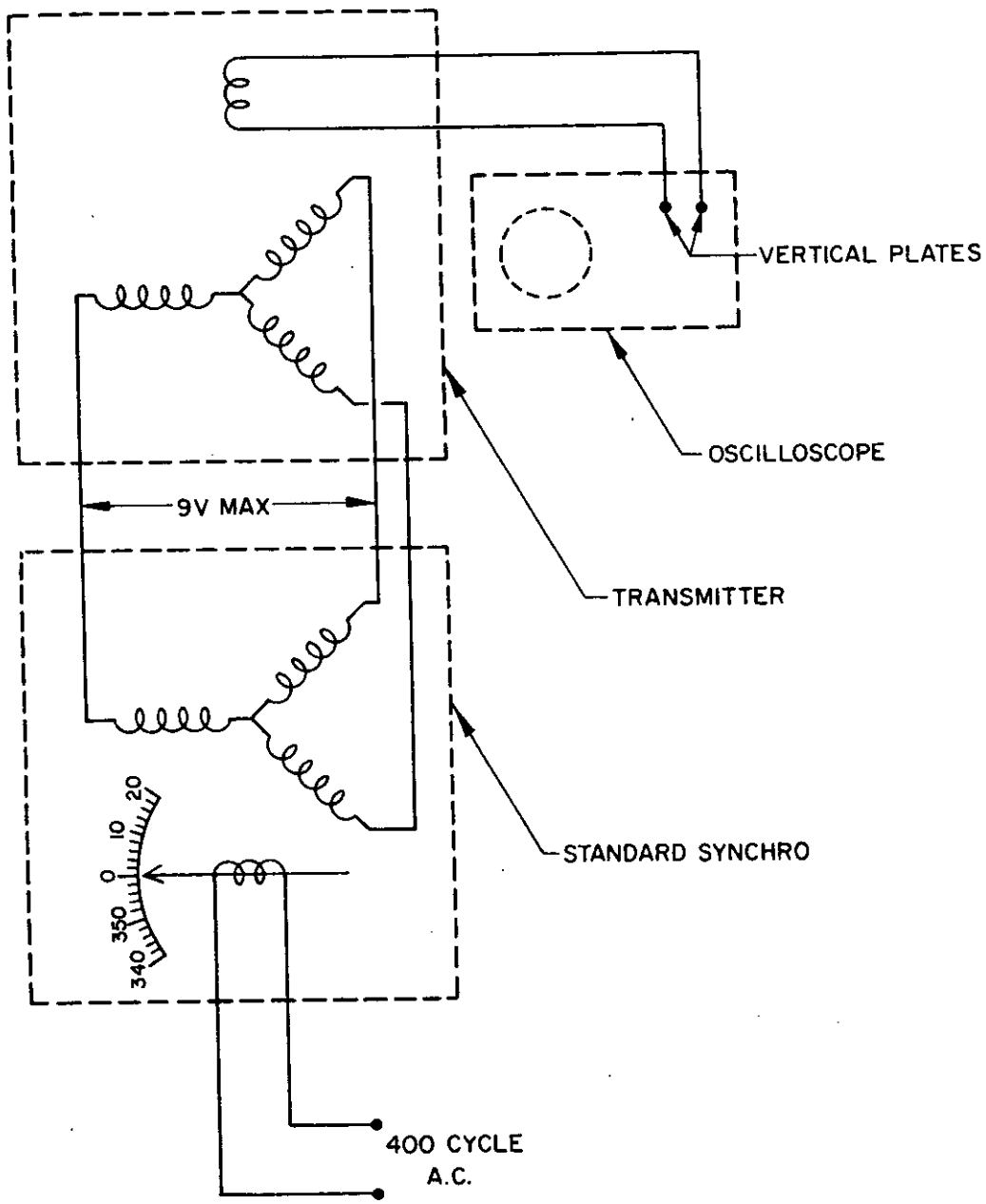


FIGURE 3. ELECTRICAL CONNECTIONS

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vibration with an amplitude between 0.009 and 0.011 inch, at frequencies from 300 to 3,000 cycles per minute, in order to determine that the natural frequency of the transmitter does not occur in this frequency range. While the transmitter is being vibrated, the maximum amplitude of the null signal shall not increase in value more than the amount specified in Table II. The angular shift of the standard synchro to maintain the null shall not be greater than the amount specified in Table II.

* 4.7.9 Position error. The indicator shall be observed while the transmitter is held in any desired position and tapped. This reading shall not differ from the reading when the transmitter is held in any other position by more than the tolerances specified in Table II. This test shall be made at every other one of the points indicated by an asterisk in Table I beginning with the first.

* 4.7.10 Vibration failure. The test shall be conducted in accordance with the vibration failure procedure V of MIL-E-5272 while the absolute pressure specified in Table II is applied to the transmitter. The difference between the indication of the pointer at the beginning and at the end of the 3 hour vibration period shall not exceed the tolerance specified in Table II, and no looseness in the mechanism or damage shall result in any part of the indicator.

* 4.7.11 Special tests. Three transmitters shall be selected at random from the first 15 transmitters representing the manufacturer's contract or order and shall be subjected to all of the following tests.

* 4.7.11.1 Environmental. The following tests shall be conducted in accordance with the specified procedures of MIL-E-5272, and as specified herein.

* 4.7.11.1.1 Low temperature exposure - Procedure III. While still at the test temperature, suction shall be applied to the static connection and electrical power to the servo system. The indicator shall operate without sticking. After the transmitter has returned to room temperature, it shall meet the requirements specified for room temperature scale error. There shall be no damage which would adversely affect subsequent operation.

* 4.7.11.1.2 High temperature exposure. Procedure I for a period of 24 hours. After the transmitter has returned to room temperature, it shall meet the requirements specified for room temperature scale error. There shall be no damage which would adversely affect subsequent operation.

* 4.7.11.1.3 Humidity - Procedure I. Upon completion of the test, the transmitter shall be inspected to ascertain that no evidence of corrosion or rusting of the internal or external parts of the transmitter has occurred as a result of this test.

* 4.7.11.1.4 Fungus - Procedure I. The transmitter shall have applicable external connections installed. After the test, the transmitter shall meet the requirements specified for individual tests. There shall be no deterioration which would adversely affect subsequent operation.

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* 4.7.11.1.5 Salt spray. Procedure I for a period of 50 hours. At the end of the 50 hour period, the individual tests shall be repeated. No damage shall result from this test and the tolerances specified for the individual tests shall apply.

* 4.7.11.2 Acceleration. The transmitter shall be mounted and accelerated first in its vertical axis and then in each of the 2 axes that are perpendicular to the vertical axis and to each other. The transmitter shall be subjected to an acceleration of 20 times the force of gravity in each of the above axes for a period of 30 seconds. At the end of this acceleration, the transmitter shall meet the requirements specified in 4.6.1.4. No damage to the transmitter shall result from this test.

* 4.7.11.3 Mounting lugs. The transmitter case with the mechanism and cap removed shall be mounted face downward on the movable head of a suitable testing machine with the diameter of the case in a horizontal plane so that the mounting lugs receive no support. A suitable pin shall be inserted through the hole in the mounting lug and attached to the pull strap in the stationary head of the machine. A load of 175 pounds shall be applied for one minute to each lug along the lug axis and toward the rear of the instrument. The lugs shall withstand the applied load without fracture.

* 4.8 Retest. All parts, specimens, or assemblies destroyed in making tests required by this specification or drawings, to determine compliance with the specification or drawings, shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of the contract or order.

* 4.9 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified (see 6.4).

* 5.1.1 Level A. Each transmitter shall be preserved and packaged in accordance with MIL-P-116, Method III and placed in a snug-fitting container conforming to Grade W6C, PPP-B-636.

* 5.1.2 Level C. Each transmitter shall be preserved and packaged in a manner which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity. The supplier may use his commercial practice providing it meets this requirement.

* 5.2 Packing. Packing shall be Level A, B or C as specified (see 6.4).

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* 5.2.1 Level A. Transmitters preserved and packaged as specified in 5.1.1 shall be packed in weather resistant shipping containers conforming to PPP-B-636. The grade of each shipping container, determined by the gross weight and size, shall comply to the applicable special requirements table of the specifications. Interior cushioning, blocking, bracing and waterproofing shall be in accordance with MIL-STD-1186. Closure and strapping shall be in accordance with the container specification. Containers shall be of uniform shape and size and of minimum cube and tare.

* 5.2.2 Level B. Same as Level A, except that waterproofing is not required and domestic containers may be used.

* 5.2.3 Level C. Packages that require overpacking for acceptance by the carrier shall be placed in exterior type shipping containers in a manner that will insure safe transportation at lowest rate to the point of delivery. Containers shall comply with Uniform Freight Classification Rules, or regulations of other carriers as applicable to the mode of transportation.

* 5.3 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

* 6.1 Use. The Type MC-1 transmitter covered by this specification is intended for use on aircraft to supply signals for operating various control equipment requiring signal in atmospheric pressure ranging from -1000 to 80,000 feet.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of packaging and packing required (see Section 5).

* 6.3 Definitions.

* 6.3.1 Amplitude. Whenever the word "amplitude" is specified, it shall mean the extent of motion as measured from one extreme to the opposite extreme.

* 6.3.2 External connections. When external connections to simulate installed conditions are called for, they shall consist of 10 feet of 1/4 inch copper or aluminum tubing coiled so that moisture can drain out the open end. Similar coils shall be connected to the pitot and static pressure connections.

* 6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded

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contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the San Antonio Air Materiel Area, Service Engineering Division, ATTN: SANE, Kelly AFB, Texas 78241, and information pertaining to qualification of products may be obtained from that activity.

* 6.5 Identification of changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 82

Review Activity:
Air Force - 82

Preparing Activity:
Air Force - 82

Project No. 6685-F324

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1426
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REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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